

Review

The Microbiome, Inflammation, and GVHD Axis: The Balance Between the “Gut” and the Bad

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Abstract: Hematopoietic stem cell transplantation is one of the most intricate immune therapies used for patients with hematological diseases or immune disorders. In addition to the inherent immunosuppression from their primary condition, many of these patients usually receive cytotoxic chemotherapy, radiation therapy, broad-spectrum antibiotics, or experience extended nutritional perturbations. These factors collectively lead to inflammation and the disruption of gut microbiota. Additionally, about 40–60% of patients undergoing fully HLA-matched allogeneic transplantation are expected to develop acute graft-versus-host disease (aGVHD), even with prophylactic measures such as calcineurin inhibitors, methotrexate/mycophenolate, or post-transplant cyclophosphamide treatment. Recent research has elucidated the complex interplay between immune effectors in the gastrointestinal tract and microbial populations within a proinflammatory peri-transplant environment, revealing its significant effect on survival and post-transplant complications such as aGVHD. This review will explore the relationship between dysbiosis during allogeneic transplantation and mechanisms that can help clarify the link between gut microbiota and the risk of GVHD, along with emerging therapeutic strategies aimed at addressing dysbiosis during hematopoietic stem cell transplantation.

Keywords: hematopoietic stem cell transplantation (HSCT); graft-versus-host disease (GVHD); microbiome; dysbiosis; GVHD regulation



Academic Editor: Toshihiko Torigoe

Received: 30 January 2025

Revised: 28 February 2025

Accepted: 4 March 2025

Published: 7 March 2025

Citation: Pinzon-Leal, P.; Gutierrez-Barbosa, H.; Medina-Moreno, S.; Zapata, J.C. The Microbiome, Inflammation, and GVHD Axis: The Balance Between the “Gut” and the Bad. *Immuno* **2025**, *5*, 10. <https://doi.org/10.3390/immuno5010010>

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1. Introduction

Hematopoietic stem cell transplantation (HSCT) is a complex procedure primarily used as a treatment for hematological diseases, both neoplastic (mainly leukemia, lymphomas, and myeloma) and non-neoplastic (such as aplasia, hereditary immune/metabolic disorders, and autoimmune diseases). Each year, approximately 20,000 hematopoietic transplants are performed in the United States and around 46,000 in Europe [1,2]. Despite being a continuously advancing therapy, the leading causes of mortality within the first 100 days remain disease relapse, infections, and graft-versus-host disease (GVHD) [3].

The procedure is based on the premise that a single pluripotent cell can regenerate a complete hematopoietic system, enabling higher-intensity cytotoxic treatments without surpassing the threshold for irreversible marrow aplasia and toxicity. There are two main types of HSCT depending on the source of the stem cells as follows: autologous transplantation (autoSCT) and allogeneic transplantation (alloSCT). AutoSCT allows for the administration of high-dose chemotherapy while avoiding prolonged aplasia by using hematopoietic cells that have been previously harvested and selected from the transplant receptor. In the case of alloSCT, the goal is to replace diseased or defective hematopoiesis

with healthy hematopoietic cells from a donor while also inducing a graft-versus-tumor (GvT) effect for curative potential for advanced hematologic malignancies [4]. In this process, donor T cells recognize the recipient's tumor-specific alloantigens, triggering an immune-mediated response that targets and eliminates residual neoplastic cells.

However, donor immune responses can also be directed against the recipient's healthy non-hematopoietic tissues, leading to GVHD. This complication remains one of the main causes of morbidity and mortality after HSCT. Donor antigen-presenting cells (APCs) present recipient alloantigens to naïve donor T cells and generate an immune response characterized by tissue damage [5]. Additional tissue damage, caused by the aggressiveness of conditioning plus chemo-radiotherapy, releases antigens and triggers an inflammatory response that facilitates the presentation to T cells and contributes to the development of GVHD. Traditionally, GVHD has been categorized based on the affected organs and the timing of onset into either acute and chronic forms. However, some patients may present with persistent, recurrent, late-onset acute GVHD or overlap syndrome [6]. Acute GVHD (aGVHD) typically manifests within the first 100 days after HSCT. It is characterized by the apoptosis of the epithelial cells, lymphocytic infiltration, and inflammatory lesions of varying severity, affecting the following three main target organs: the skin, liver, and gastrointestinal (GI) tract. Clinical manifestations of aGVHD generally include skin rashes, nausea, abdominal pain, diarrhea, and elevated bilirubin levels. These features are relatively non-specific and can overlap with other conditions, such as infections and drug toxicity, making the diagnosis reliant on clinical and histologic criteria while excluding other potential causes [7,8]. Over time, chronic GVHD (cGVHD) may develop, with clinical manifestations that can involve almost any organ and resemble those of systemic autoimmune diseases (such as Sjögren's syndrome and scleroderma). This organ damage is due to fibroblast activation and chronic inflammation, predominantly affecting the skin, mouth, eyes, genitalia, esophagus, lungs, liver, muscles, and fascia [9].

Different authors have identified the digestive tract as an initial niche for alloreactivity, as various antigen-presenting cells (APCs), such as dendritic cells, mesenchymal cells, and intestinal epithelial cells, can contribute to the activation and differentiation of the donor's pathogenic T cell population [10]. Naïve T cells can rapidly infiltrate the GI tract within 6 h of transplantation; and non-hematopoietic APCs in the GI tract can interact with T cells, leading to activation within 24 h [11]. An inflammatory milieu is created by the disruption of the intestinal barrier caused by chemotherapy/radiotherapy during conditioning, allowing the release of pathogen-associated molecular patterns (PAMPs) from the local microbiome and damage-associated molecular patterns (DAMPs) from damaged tissue [12]. The PAMP/DAMP signal contributes significantly to the activation and proliferation capacity of dendritic cells (DCs) in the GI, allowing for better alloantigen presentation, migration to mesenteric lymph nodes, the amplification of the T cell response, and the induction of the $\alpha 4\beta 7$ receptor and other integrins which provide homing signals to the intestine to perpetuate the GI tract aGVHD [13].

Approximately 10–100 trillion microorganisms coexist symbiotically in each human, forming what we call the microbiota. Around 95% of the microbiota are in the gastrointestinal tract, with approximately 90% present in the colon. The "microbiome" refers to the microbiota, their genes, metabolites, and specific environment [14,15]. Numerous studies in both animal models and humans have highlighted the central role of the gut microbiome in maintaining immunity, metabolic balance, and neurohormonal regulation [16]. An imbalance in the microbiota is referred to as dysbiosis, and it can be defined as a functional and/or compositional alteration of the microbiota in diseased individuals compared to healthy subjects [17]. The administration of intensive chemotherapy, radiotherapy, the use of antibiotics, gastrointestinal infections (e.g., *Clostridium difficile*), dietary changes, and

nutritional disturbances are some of the factors contributing to dysbiosis prior to transplantation, wherein reductions in fecal α -diversity have been described [18]. Post-transplant conditioning regimens, prolonged exposure to broad-spectrum antibiotics during prophylaxis, and neutropenia are associated with the deterioration of diversity in intestinal microbiota and the decline in healthy commensal microbes (primarily composed of members from the Firmicutes and Bacteroidetes phyla, along with populations of Proteobacteria, Actinobacteria, Verrucomicrobia, and Fusobacteria), as well as an expansion and dominance of certain species such as *Enterococci*, *Streptococci*, and Proteobacteria [19].

Many studies and research groups have linked microbiota to the risk of developing GVHD. However, differences in practices for performing hematopoietic stem cell transplants, as well as patient characteristics (such as cancer type, prior chemotherapy/radiotherapy, and nutritional status), transplant types (cell source and donor mismatch), treatment protocols (conditioning regimen, T cell depletion platform, GVHD prophylaxis, and use of prophylactic antibiotics), and other unidentified factors, lead to significant heterogeneity across the findings from these studies [20–31]. In this review, we identify the common patterns across studies, relevant factors with biological plausibility, and mechanisms that can help clarify the link between the microbiota and the risk of GVHD.

2. Pre-Clinical Studies

Since the 1970s, studies using animal models have shown that germ-free mice undergoing bone marrow transplantation exhibited higher survival rates and an apparent absence of GVHD compared to mice without decontamination [32]. However, these “germ-free” mice presented a series of immunological alterations, such as underdeveloped spleens and lymph nodes, reduced lymphocyte counts, and other alterations in cellular populations that could interfere with proper immune function and the development of “secondary disease”, a term that we now recognize as GVHD [33].

Subsequent studies, using animal models in mice and dogs treated with antibiotics as a decontamination method, also found lower rates of GVHD and increased survival rates [34–37]. However, in mice treated with ampicillin, a reduction in *Lactobacillus* species was associated with worse GVHD outcomes, while the reintroduction of *L. johnsonii* protected against the dominance of *Enterococcus* species and reduced GVHD lethality [20]. These findings shed light on the profound impact of the gut microbiome on HSCT outcomes and how different types of microorganisms could modulate the severity of GVHD in animal models and, by extrapolation, in humans.

3. Clinical Studies

In 2012, Jenq et al. demonstrated similar findings between murine models and a cohort of transplanted patients with GVHD. They found that this population, unlike patients who did not develop any clinical signs of GVHD, exhibited lower microbiota diversity, an increase in *Lactobacillus* species, and a decrease in *Clostridiales* levels [20]. Over time, other research groups sought to demonstrate microbiota changes in HSCT patients and to establish possible associations with transplant outcomes (Table 1).

Table 1. Clinical studies investigating microbiota diversity and its impact on various outcomes in hematopoietic stem cell transplantation (HSCT).

Patient No, Population	Pathology	HSCT Characteristics ⁺	Antimicrobial Prophylaxis	Microbiota Findings	Outcomes	Ref
<i>n</i> = 18, adults	AML/MDS/MPN 72%, NHL 28%	MAC 22%, RIC 56%, NMA 22% PBSCs 33%, UCB 67%	Vancomycin, fluoroquinolone, metronidazole	- Diversity lost over time, particularly after GVHD. - GVHD profile: ↑ Lactobacillales and ↓ Clostridiales.	Increased microbial “chaos” early after allogeneic BMT as a potential risk factor for subsequent GVHD.	[20]
<i>n</i> = 80, adults	Leukemia 31%, lymphoma 42%, MM 8%, MDS 19%	- MSD 42%, MUD 42%, MMUD 8%, UCB 7% - MAC 42%, RIC 27%, NMA 31% - T cell depletion (ex vivo) 42%	Ciprofloxacin and IV vancomycin if viridians streptococci ++	- Low diversity microbiota associated with worse survival and TRM. - Dominance during HSCT: <i>Enterococcus</i> , <i>Streptococcus</i> , <i>Enterobacteriaceae</i> , and <i>Lactobacillus</i>	OS 3y: Low 36%, intermediate 60%, and high diversity 67% (<i>p</i> = 0.019).	[21]
<i>n</i> = 31, adults	AML 45%, ALL 16%, Ly 13%, MM 9%, SMD 19%, CML 3%, AA 3%	- MSD 42%, MUD 58% - MAC 35%, RIC 65%	Ciprofloxacin and metronidazole	- During HSCT microbiota shift: ↑ <i>Enterococci</i> and ↓ other <i>Firmicutes</i> (especially if AB use or GI aGVHD). - GVHD profile: ↑ <i>Enterococci</i> .	- Indoxyl sulfate levels lower during AB and in GI GVHD (indirect marker of bacterial diversity).	[22]
<i>n</i> = 64, adults.	NHL 37%, AML 38%, ALL 10%, HL 5%, CLL 6%, MDS 4%	- MSD 27%, MUD 25%, MMUD 48% - MAC 19%, RIC 50%, NMA 31% - In vivo T cell depleted with ATG	Ciprofloxacin and IV vancomycin ++	- Increased bacterial diversity was associated with reduced GVHD lethality. - ↓ <i>Blautia</i> associated with anti-anaerobic AB and parenteral nutrition.	- ↑ <i>Blautia</i> associated with reduced GVHD mortality, incidence of aGVHD, and improved OS.	[23]

Table 1. Cont.

Patient No, Population	Pathology	HSCT Characteristics ⁺	Antimicrobial Prophylaxis	Microbiota Findings	Outcomes	Ref
<i>n</i> = 15, children	SCA 27%, AML 20%, ALL 13%, HLH 13%, DBA 7%, AA 7%, FA 7%, SCID 7%	- Bone marrow 93%, CBU 7% - MSD 47%, MUD 27%, MMUD 27% - MAC 60%, RIC 40%	Levofloxacin prophylaxis (53%)	- GVHD profile: ↑ <i>Enterococcus</i> spp. and <i>Enterobacteriaceae</i> family but ↓ anti-inflammatory <i>Clostridia</i>.	- Cumulative Ab exposure associated with GVHD (OR 1.11 <i>p</i> < 0.008).	[24]
<i>n</i> = 107, adults	ALL 16%, AML 46%, MDS 21%, other 17%	- PBSC 28%, BM 61%, CBU 11% - MAC 61%, RIC 39% - HLA mismatch: 0 (50%), 1 (28%), 2 (9%) >= 3 13%	- Levofloxacin and itraconazole x14d before HSCT ++	- Stool samples collected 2 weeks before conditioning. Microbiota diversity classified as low, intermediate, or high. - GVHD profile: ↑ <i>Firmicutes</i> (<i>p</i> < 0.01) and ↓ <i>Bacteroides</i> (<i>p</i> = 0.106).	- No difference in OS, relapse, NRM, or GI GVHD II-IV incidence between diversity groups.	[25]
<i>n</i> = 66, adults	ALL 6%, AML 26%, CLL 6%, HL 3%, MDS 36%, MM 3%, NHL 15%	- Matched 90% - NMA 31%	Not reported	- GVHD associated with lower alpha diversity. - GVHD profile: ↑ oral <i>Actinobacteria</i> and ↑ <i>Firmicutes</i> in stool, ↓ <i>Lachnospiraceae</i>.	- Stool microbiota around the time of neutrophil recovery are predictive of severe aGVHD.	[26]
<i>n</i> = 541, adults	AML 36%, MDS 16%, NHL 17%, MM 11%, ALL 8%, other neoplasia 17%	MAC 59%, RIC 30%, NMA 11% PBSCs/BM 32%, CBU 18%, T cell depletion 50%. Donor related 30%, unrelated 70%	Not reported	- Stool samples collected after HSCT and 2 y of follow up. - GI diversity not associated with relapse/progression.	- Presence of <i>Eubacterium limosum</i> decreased risk of relapse/progression (HR 0.82, <i>p</i> = 0.009) and increase OS, but not associated with aGVHD.	[27]

Table 1. Cont.

Patient No, Population	Pathology	HSCT Characteristics ⁺	Antimicrobial Prophylaxis	Microbiota Findings	Outcomes	Ref
<i>n</i> = 141, adults	AML 60%, ALL 35%, MDS 5%	- MSD 45%, Haplo 50%, MUD 5% - High intensity conditioning 43%	Oral sulfamethoxazole and norfloxacin	- aGVHD profile: lower diversity ($p = 0.018$) and ↓ Clostridia (e.g., the Lachnospiraceae and Ruminococcaceae families) and ↑ Gammaproteobacteria (e.g., the Enterobacteriaceae family). - Higher conditioning intensity and b-lactam AB associated with lower diversity and Treg/Th17.	- Accumulated intestinal microbiota correlated with aGVHD grade and could predict development of aGVHD.	[28]
<i>n</i> = 1324, adults	AML 36%, MDS/MPN 18%, NHL 17%, ALL 9%, MM 9%, CLL 3%, CML 2%, HL 2%, AA 1%, Other 3%	- BM/PBSC 50%, CBU 15%, PBSC T cell depleted 35% - MAC 56%, RIC 35%, NMA 9%	Not reported	- Depletion of lactose in vitro and in vivo inhibited <i>Enterococcal</i> expansion and mitigated GVHD. - GVHD profile: ↑ <i>Enterococcus</i> domination.	- <i>Enterococcus</i> domination increases risk for GVHD and overall GVHD-related mortality. - Lactose-free diet could attenuate the outgrowth of pathobionts	[29]
<i>n</i> = 1362, adults	AML 36%, MPN/MDS 19%, NHL 17%, ALL 9%, MM 8%, CLL 2%, HL 2%, CML 2%, AA 1%, other 3%.	BM 8%, CBU 15%, PBSCs unmodified 43%, PBSCs T cell depleted 33% MAC 57%, RIC 34%, NMA 9%	- Cipro/levofloxacin; one center used rifaximin	- Before HSCT samples already showed microbiome disruption. - HSCT causes loss of diversity ($p < 0.001$) and domination of a single taxa.	- Lower diversity associated with lower OS (HR 0.75), TRM (HR 0.63), and GVHD-related mortality (HR 0.89). - Lower mortality if higher diversity at time of Neutrophil engraftment.	[30]

Table 1. Cont.

Patient No, Population	Pathology	HSCT Characteristics +	Antimicrobial Prophylaxis	Microbiota Findings	Outcomes	Ref
n = 74, children	AML 59%, ALL 39%, HL 3%, LDS 3%, other neoplasia 3%.	Haplo 61%, MUD 26%, MSD 12%, MMUD 1% BM 36.5%, PBSCs 63.5% RIC 67%, MAC 33%, NMA 1% T cell depletion ex vivo 51.4%	No oral decontamination or antibacterial prophylaxis peri-HSCT.	- Microbiome indices described in adults do not predict outcomes in children.	- Low diversity or low butyrogens not associated with clinical outcomes.	[31]
					- Facultative anaerobes (e.g., <i>Lachnoclostridium</i> , <i>Parabacteroides</i>) prior to conditioning predicted bacteremia risk (HR 3.89). - Compositional signatures predict both infectious and aGVHD.	

Abbreviations: Myeloablative conditioning (MAC), reduced intensity conditioning (RIC), nonmyeloablative conditioning (NMA). Matched sibling donor (MSD), matched unrelated donor (MUD), miss-matched unrelated donor (MMUD), umbilical cord blood (UCB), bone marrow graft (BM), peripheral-blood stem cells (PBSCs). Antibiotic prophylaxis used routinely (R). Overall survival (OS), transplant-related mortality (TRM). Non-Hodgkin lymphoma (NHL), acute myeloid leukemia (AML), acute lymphoblastic leukemia (ALL), Hodgkin lymphoma (HL), chronic lymphocytic lymphoma (CLL), myelodysplastic syndrome (MDS), multiple myeloma (MM), aplastic anemia (AA), sickle cell anemia (SCA), hemophagocytic lympho-histiocytosis (HLH), Diamond–Blackfan anemia (DBA), Fanconi anemia (FA), severe combined immunodeficiency (SCID). Anti-thymocyte globulin (ATG), antibiotics (AB). + HSCT characteristics refer to type of HLA matching, stem cell source, conditioning intensity, and possible T cell depletion (ex vivo or in vivo platforms). ++ Use of trimethoprim-sulfamethoxazole, aerosolized pentamidine, or atovaquone for *P. jirovecii* prophylaxis. ↑: Increase, ↓: Decrease.

The outcomes of these clinical studies are varied, largely due to differences in patient populations, hematologic disease, and transplant protocols across studies. Some studies use grafts with ex vivo T cell depletion [21,27,29], which has shown a significant reduction in the risk of GVHD, but it was also shown to impair post-transplant immune recovery and it may alter the potential for a GvT effect. Interestingly, Peled's study investigated the impact of gut microbiota diversity in patients with T cell-depleted versus T cell-replete grafts, showing that, in the latter group, higher microbiota diversity was associated with better overall survival (OS), lower rates of transplant-related mortality (TRM), and reduced GVHD-related mortality. In contrast, the cohort with T cell-depleted grafts showed no association between microbiota diversity and transplant outcomes [27]. It is likely that mature T cells from the graft are needed to generate early alloimmune reactivity with microbiota during transplantation, though other factors related to the conditioning regimens and GVHD prophylaxis in these two cohorts cannot be ruled out.

The conditioning regimens and their intensity vary significantly among groups, leading to differences in the impact on the microbiota. More intense protocols, including total body irradiation (TBI), thiopeta, or cyclophosphamide, are associated with a greater loss of microbiota diversity [38]. Antimicrobial prophylaxis protocols that include anti-anaerobic antibiotics and the use of intravenous vancomycin during transplantation alter microbiota diversity early on. Early exposure to these antibiotics is associated with lower levels of urinary 3-indoxyl sulfate (indicative of a disrupted gut microbiome) and a reduction in commensal Clostridiales species [39].

Despite the various factors that can influence the analysis and comparability of studies, some findings remain consistent among groups. In most studies, a reduction in commensal flora and the dominance of a single species lead to decreased intestinal microbiota diversity, which is associated with worse transplant outcomes, such as an elevated risk of GVHD and increased mortality. Another key finding is that the composition of bacterial species and their function are crucial, although due to the high diversity of stool bacteria and the complexity of their taxonomy, it is difficult to identify a single "culprit" microorganism. However, the expansion of *Enterococcus* has been linked to bloodstream infections, GVHD, and mortality in most studies (Table 1).

4. Antibiotics Impact in Microbiome

One of the most extensively studied factors affecting the microbiota in HSCT patients is the detrimental impact of antibiotics, which are commonly used in this population in the following two main scenarios: as prophylaxis, with antibiotics routinely started during conditioning to prevent infections in the pre-engraftment period; and as empiric therapy, involving the use of broad-spectrum antibiotics in cases of neutropenic fever due to aplasia and prolonged immunosuppression.

A recent survey conducted across various transplant centers of the European Society for Blood and Marrow Transplantation (EBMT) revealed that 69% of transplant centers routinely administered antibacterial prophylaxis in allo-HSCT, primarily with quinolones (e.g., levofloxacin). However, the duration and type of antibiotic varied significantly among centers [40]. Antibacterial prophylaxis is used as a gut decontamination strategy to minimize the bacterial load that could be translocated due to mucositis and aplasia, thereby reducing the risk of bloodstream infections [41]. However, the use of prophylactic antibiotics is controversial since some studies have linked the use of prophylactic antibiotics with the increase in multidrug-resistant bacteria [42,43], *Clostridium difficile* infections [44], increased rates of aGVHD [45,46], and the increase in drug interactions and toxicities (hepatotoxicity, peripheral neuropathy, tendon rupture, and QT prolongation) [41,47]. The use of non-absorbable oral antibiotics like rifaximin and protective environments (laminar airflow

isolation and sterile or low bacteria food) has shown delays in the onset and a reduction in the incidence of GVHD in patients undergoing HLA-matched sibling bone marrow transplants for aplasia [48–50]. However, no impact on GVHD or survival rates was observed in patients with various hematological malignancies using prophylactic antibiotics such as quinolones [51,52]. The Essen Transplant Group conducted a randomized decontamination study using ciprofloxacin, with or without metronidazole, and found a significant reduction in aGVHD (grades II–IV) incidence in the ciprofloxacin + metronidazole group (25% vs. 50%, $p < 0.002$) in patients with myeloid malignancies undergoing HLA-matched sibling transplants, with no differences in cGVHD or survival rates. They also demonstrated the suppression of anaerobic bacterial growth for 5 weeks post-transplantation [53,54]. Other groups have been unable to replicate the results of the pre-HSCT decontamination strategy, and on the contrary, have found that the use of anti-anaerobic antibiotics reduces anti-inflammatory Clostridia and increases the incidence of aGVHD [24,45]. Recently, a study evaluated 1214 patients who received antibiotics with an anti-anaerobic spectrum activity, and its use was associated with higher risk of aGVHD of gut/liver (HR 1.26) and aGVHD mortality (HR 1.63), lower gut bacterial diversity by the metagenomic sequencing of serial fecal samples that specifically found a reduction in *Bifidobacteriales* and *Clostridiales*, along with a loss of bacterial genes related with butyrate synthesis [55]. Gavrilaki and colleagues conducted a systematic review and meta-analysis examining the association between the antibiotic-mediated disruption of the microbiota and key allo-HSCT outcomes. The main findings supported the conclusion that an increased aGVHD risk is associated with the use of antibiotic prophylaxis (OR 1.65) and gut decontamination strategies (OR 1.58). Additionally, patients with higher microbiota diversity had better survival rates and lower transplant-related mortality [56]. The routine use of prophylactic antibiotics has not been shown to impact the survival or infection-related mortality in HSCT [57], and their use is now reserved for individual cases following a careful risk-benefit evaluation, in line with the clinical guideline recommendations (recommendation against use: ECIL-8 [58] and IDSA 2020 [59] for pediatric patients; recommendation to use fluoroquinolone prophylaxis with low-level evidence: ASCO/IDSA 2018 [60], NCCN 2024 [61]).

Approximately 80% of HSCT recipients will develop febrile neutropenia, and in only 30% of cases will the infectious etiology be identified [62]. All clinical guidelines emphasize the critical need to urgently start empirical broad-spectrum antibacterial therapy within the first hour of fever (IDSA 2010 [63], ECIL 2011 [64]) since delay could lead to increased mortality in this high-risk patient group [65]. The cornerstone of antimicrobial therapy consists of intravenous broad-spectrum β -lactam antibiotics with anti-pseudomonal activity (e.g., piperacillin/tazobactam or cefepime). In cases of multidrug-resistant (MDR) bacteria, a carbapenem should be considered. There are also specific recommendations for using aminoglycosides, broad-spectrum anti Gram-positive therapy, and new combination antimicrobials [64,66]. Several studies have highlighted the association between antibiotic use and aGVHD, as well as GVHD-related mortality, particularly with antibiotics that have an anti-anaerobic activity, such as piperacillin/tazobactam and imipenem/cilastatin [46,67,68]. A recent analysis of a cohort of 2023 allo-HSCT patients from the Fred Hutchinson Center found that exposure to carbapenems and piperacillin/tazobactam during the first two weeks after allo-HSCT was associated with the highest risk of subsequently developing aGVHD [69]. In a study by Nørgaard, 577 fecal samples from heavily antibiotic-treated HSCT patients were examined using metagenomics. The most used antibiotics included TMP/SMX, quinolones, amoxicillin/clavulanic acid, meropenem, clindamycin, piperacillin/tazobactam, and vancomycin IV. The study found that broad-spectrum antibiotic treatment with meropenem or piperacillin/tazobactam was associated with the largest

decrease in bacterial diversity and an increase in antibiotic-resistant genes, particularly the selection of *E. faecium* and the *vanA* and *vanC* genes [70].

5. Molecular Microbiological Techniques and Their Impact on Microbiota Studies in HSCT

Advancements in molecular microbiological techniques, particularly 16S ribosomal RNA (rRNA) polymerase chain reaction (PCR) sequencing, and metagenomic and metatranscriptomic analyses have made it possible to identify, classify, and quantify microorganisms within the intestinal microbiome [71]. One of the most widely used methods for defining microbial diversity is based on the 16S rRNA analysis. This involves extracting DNA from fecal biospecimens, amplifying the V4-V5 regions of the 16S ribosomal RNA using modified universal bacterial primers, and then sequencing the purified products to compare them with known taxonomic units [21]. Microbial diversity can be quantified using indices such as the inverse Simpson index and the Shannon diversity index [72,73]. Among the approximately 100 trillion bacteria that make up the gut ecosystem, three main phyla—Bacteroidetes, Firmicutes, and Actinobacteria—comprise more than 90% of the dominant microbiota in healthy individuals. However, next-generation sequencing (NGS) studies have revealed significant diversity and variability at lower taxonomic levels between individuals, with each person carrying at least 160 different species [74]. Research has shown that shifts in microbiome diversity and the dominance of specific bacteria are associated with key outcomes in transplantation. A decrease in the abundance of Clostridia, Lachnospiraceae, Blautia, Bacteroides, and *Akkermansia muciniphila* is linked to a higher probability of acute graft-versus-host disease (aGVHD) and lower survival rates [75,76]. Conversely, the abundance of Lactobacillales, Staphylococcaceae, Enterobacteriales, and Enterococcus has been observed in patients with aGVHD and is concomitantly associated with GVHD severity, lower overall survival, and higher GVHD-related mortality [22,77].

The impact of microbiome diversity on HSCT outcomes has been evaluated in single-center studies. Recently, a multicenter study conducted across four transplant centers profiled 8767 fecal samples, obtained from 1362 allo-HSCT recipients. The microbiota composition was analyzed through ribosomal RNA gene sequencing at various time points during the transplantation process. The study revealed a decline in microbiota diversity over HSCT time. Interestingly, samples from all four centers demonstrated the common dominance of the genera Enterococcus and Streptococcus. Lower microbiome diversity was associated with higher rates of transplantation-related mortality and GVHD-related mortality [78].

6. Mechanism of Action Between the Microbiome and GVHD

There are multiple interactions between the microbiome and the immune system that can regulate GVHD. These include immune system modulations, such as interactions with the microbiome-gut-associated lymphoid tissue (GALT) and the regulation of T cell activation; the production of metabolites such as short-chain fatty acids (SCFAs) and indoleamine 2,3-dioxygenase (IDO); the intestinal barrier function of the microbiome; inflammation; and cytokine production leading to dysbiosis [79–81].

Acute GVHD is characterized by a strong inflammatory response derived by Th1 and Th17 immune response [82]. The crosstalk between different immunological pathways is responsible for this profile. In this section, we review how the microbiome can regulate the GVHD immunological profile through the activation of T cells in a HLA-dependent manner, the Toll-like receptor (TLR) response, and the activation of the inflammasome.

6.1. HLA Regulation by the Microbiome During GVHD

Graft-versus-host disease is the result of disparities in the Human Leukocyte Antigen (HLA) between the recipient and the donor. This mismatch is presented conformationally as an alloantigen to the donor’s CD8+ and CD4+. The incidence of GVHD is higher in transplants with a greater degree of HLA mismatch (e.g., >8/10), although, even in transplants from HLA-identical donors (10/10 similarity), differences in minor histocompatibility antigens (mHLA) can still lead to allorecognition and the development of GVHD [83].

Allorecognition can occur through the following three main pathways: the direct pathway, in which donor antigen-presenting cells (APCs) present donor major histocompatibility complex (MHC) molecules directly to the recipient T cells; the indirect pathway, where the recipient APCs process and present allopeptides via their own MHC molecules to the recipient T cells; and the semi-direct pathway, where the recipient APCs acquire donor MHC molecules bound to allopeptides and present them to the recipient T cells [84]. All these pathways highlight the critical role of MHC presentation and its contribution to the activation of various T cell subsets during GVHD. CD8+ T cells and CD4+ T cells—particularly the Th1 and Th17 subsets—play pivotal roles in the progression of the disease [85,86].

Studies in mice have shown that the gut microbiota can either upregulate or downregulate the expression of MHC class II, directly influencing the severity of GVHD. Koyama et al. demonstrated that interleukin-12 (IL-12) secreted by lamina propria lymphocytes, in a microbiota-dependent manner, induces MHC class II expression on intestinal epithelial cells (IECs). This process facilitates the priming of donor CD4+ T cells, which subsequently leads to the development of lethal acute GVHD. Moreover, researchers have found that conditioning with a total body irradiation protocol in mice enhances IL-12 and interferon-gamma (IFN-γ) secretion by conventional T cells, which results in the overexpression of MHC class II on IECs [79].

In another study, genetically identical BALB/c mice from different vendor sources exhibited distinct intestinal microbiota, enabling the identification of bacterial MHC-II inducers and MHC-II suppressors on IECs in both fecal and ileal samples (Tables 1 and 2). Bacteria that induce MHC-II in the ileum promoted the expansion of T cells that secrete interferon-gamma (IFN-γ), while MHC-II-suppressing bacteria inhibited the expansion of IFN-γ-producing T cells. Notably, most bacteria that induce MHC-II were sensitive to vancomycin. The depletion of these bacteria with vancomycin not only reduced MHC-II expression and IFN-γ production but also decreased the lethality of GVHD [87].

Table 2. Fecal MHC-II-inducing and -suppressing bacteria in genetically identical BALB/c mice from different vendor origins.

Vendor	MHC-II Inducer	MHC-II Suppressor	Ref
JAX Mice	• <i>Muribaculaceae</i>	• <i>Pseudoflavonifractor</i>	[79]
	• <i>Lactobacillus animalis/apodemi/murinus</i>	• <i>Acetatifactor muris</i>	
		• <i>Clostridium celatum/saudiense</i>	
		• <i>Hungatella</i>	
		• <i>Bacteroides thetaiotaomicron</i>	
		• <i>Clostridium</i>	
		• <i>Lachnospiraceae/Ruminococcaceae</i>	
		• <i>Clostridium celatum/disporicum/saudiense</i>	
		• <i>Adlercreutzia</i>	
		• <i>Paramuribaculum</i>	
		• <i>Roseburia</i>	

Table 2. Cont.

Vendor	MHC-II Inducer	MHC-II Suppressor	Ref
Taconic and Charles River	• <i>Muribaculaceae</i> • <i>Lactobacillus animalis/apodemi/murinus</i> • <i>Candidatus arthromitus</i> • <i>Lactobacillus reuteri</i> • <i>Prevotella</i> • <i>Faecalibaculum rodentium</i>	• <i>Acetatifactor muris</i> • <i>Clostridium celatum/saudiense</i> • <i>Hungatella</i> • <i>Clostridium</i> • <i>Adlercreutzia</i> • <i>Roseburia</i>	[79]

These findings confirm that the microbiota directly influence GVHD outcomes, where suppressor bacteria will inhibit the activation of T cells in a MHC-II direct pathway, directly impacting the Th1 and Th17 T cell population, which will reduce the production of proinflammatory cytokines like IFN- γ , a cytokine that promotes the apoptosis of intestinal epithelia crypt cells and acts synergistically with LPS to stimulate the production of more cytokines (Figure 1) [88].

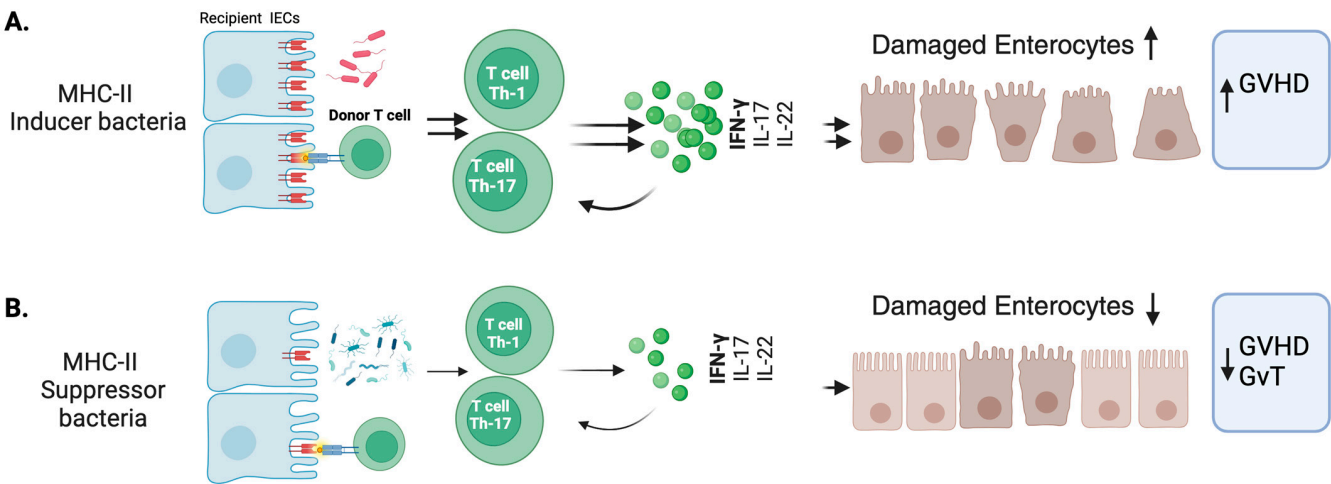


Figure 1. MHC-II-inducing and -suppressing bacteria in GVHD. (A) MHC-II-inducing bacteria upregulate the expression of MHC-II on IECs, leading to the activation of T cells, which polarize into the Th1 or Th17 subsets. These T cell populations release proinflammatory cytokines such as IFN- γ , IL-17, and IL-22, which increase enterocyte damage and contribute to severe gastrointestinal GVHD. (B) MHC-II-suppressing bacteria downregulate the expression of MHC-II on IECs, reducing T cell activation and the expansion of the Th1 and Th17 populations. This results in a lower cytokine release, which decreases enterocyte damage, reduces the severity of GVHD, and has an impact on the GvT. GVHD: graft-versus-host disease; IECs: intestinal epithelial cells; GvT: graft-versus-tumor. Created in BioRender.com.

An important finding in the regulation of MHC-II by the microbiome is its impact on the GvT response, which holds significant therapeutic potential in the treatment of advanced hematological malignancies. Lethally irradiated mice, transplanted with bone marrow and CD4+T cells with primary B cell acute lymphoblastic leukemia, were treated with a three-antibiotic cocktail (cefoxitin, gentamicin, and vancomycin) to increase the MHC-II suppressor bacteria. The treatment not only reduced the GVHD but also attenuated the GvT [87]. Together, this highlights the complex balance of bacterial populations in gastrointestinal tissue and underscores the importance of prioritizing the desired antineoplastic effect when approaching treatment in patients with specific hematological pathologies.

In addition to regulating MHC-II, the microbiome can shape the immune response in ways that may impact the development of GVHD to some extent. The commensal bacterium *Bacteroides fragilis* persists in the gastrointestinal tract by establishing a host–microbial symbiosis. To achieve this, its polysaccharide A activates TLR2 on Treg cells, inducing their proliferation and the induction of immunological tolerance [89]. This response could mitigate GVHD to some degree. On the other hand, filamentous bacteria in mice have been shown to increase the production of IL-17 and IL-22 by Th17 cells in the lamina propria [90]. As mentioned earlier, the Th17 response is one of the immune pathways involved in the severity of GVHD. This implies that the composition of the microbiota not only modulates the immune response through the upregulation and downregulation of HLA but also by favoring either immunological tolerance or a proinflammatory state, which can promote the development of severe GVHD in patients. However, further studies are needed to understand the optimal microbiota composition that could drive an immune response preventing GVHD (Table 3).

Table 3. Ileal MHC-II-inducing and -suppressing bacteria in genetically identical BALB/c mice from different vendor origins.

Vendor	MHC-II Inducer	MHC-II Suppressor	Ref
JAX Mice	• <i>Lactobacillus animalis/apodemi/murinus</i> • <i>Muribaculaceae</i>	• <i>Clostridium celatum/saudiense</i> • <i>Adlercreutzia</i> • <i>Lactobacillus gasseri</i> • <i>Paramuribaculum</i> • <i>Clostridium celatum/disporicum/saudiense</i> • <i>Romboutsia ilealis</i> • <i>Parasutterella excrementihominis</i> • <i>Olsenella</i> • <i>Clostridium</i>	[79]
Taconic and Charles River	• <i>Candidatus arthromitus</i> • <i>Lactobacillus reuteri</i> • <i>Faecalibaculum rodentium</i> • <i>Lactobacillus animalis/apodemi/murinus</i> • <i>Muribaculaceae</i>	• <i>Clostridium celatum/saudiense</i> • <i>Lactobacillus gasseri</i> • <i>Paramuribaculum</i> • <i>Parasutterella excrementihominis</i>	[79]

6.2. Toll-like Receptors' Regulation by the Microbiome in GVHD

Besides the impact of MHC-II expression and the subsequent activation of T cells, during radiation and chemotherapy conditioning, the epithelia of GI are damaged, allowing the translocation of microbiota across the mucosal barrier and the alteration of the microbiota composition, which in mice has been confirmed to affect post-irradiation survival [91,92]. These translocation events and the changes in the microbiota expose the APCs to different danger-associated molecular patterns (DAMPs) that activate Toll-like receptors (TLRs). Even though the expression of TLR 1-9 has been described in IECs, it is important to mention that low levels of TLR2 and TLR4 in combination with the abundant expression of TLR3 and TLR5 are a characteristic profile in the colon under a normal microbiome environment [93–95]. The use of conditioning protocols with 5-fluorouracil (5-FU), methotrexate (MTX), and irinotecan (CPT-11) has shown an increase in TLR expression, especially TLR-2 and -4 in 5-FU or MTX [96,97]. However, in vitro studies showed a preferential enhanced TLR-4 when Caco-2 cells were exposed to 5-FU in combination with LPS [98]. On the other hand, conditioning with CPT-11 reduces the expression of TLR-5 [99]. This demonstrates how the conditioning protocol disrupts the normal expression profile of TLRs in the gastrointestinal tissue.

The changes in TLR profile and activation can also enhance the healing process in a GVHD proinflammatory environment. Even though TLR-2 was overexpressed under different conditioning protocols, as described previously, the activation of this receptor has been reported to enhance the repair of intestinal epithelial cells (IECs), including improving barrier function, cell survival, intercellular communication, and tissue restitution, while reducing proinflammatory immune responses in both acute and chronic murine colitis [100,101]. In this context, the use of *Lactobacillus* in C57BL/6 before whole-body irradiation reduces irradiation-induced epithelial injury in mice through the activation of the TLR and the posterior induction of COX-2 in mesenchymal stem cells that preserve epithelial proliferation [102]. Another example that shows TLR activation can reduce the GVHD is TLR-5. It has been shown that the use of flagellin as an agonist of TLR-5 ameliorates GVHD in lethally irradiated mice [103]. However, the mechanism of action is still unknown.

The TLR-MyD88 activation pathway can also trigger the production of the regenerating family member gamma (Reg3 γ) in mice or the Reg3 α in humans [104]. This protein is abundant in the gastrointestinal tract under normal conditions, where it serves as an antimicrobial peptide, preventing bacterial translocation and regulating intestinal inflammation [105,106]. In a GVHD context, this protein acts as a survival signal for intestinal stem cells (ISCs) and Paneth cells, and its levels decrease in the gastrointestinal tract as GVHD progresses [107]. The treatment with IL-22 or the use of specific bacteria such as *Bifidobacterium breve*, *Lactobacillus rhamnosus* GG, or probiotics containing multiple strains of *Lactobacillus* and *Bifidobacterium* spp. appears to increase Reg3 γ expression in the intestine, which can inhibit GVHD progression [107]. It is important to note that no impact on the graft-versus-leukemia (GVL) effect has been found with the activation of these pathways.

Evidence suggests that the reduction in GVHD through the microbiota can be achieved by maintaining the integrity of the intestinal epithelium, which reduces bacterial translocation without affecting the GvT response (Figure 2).

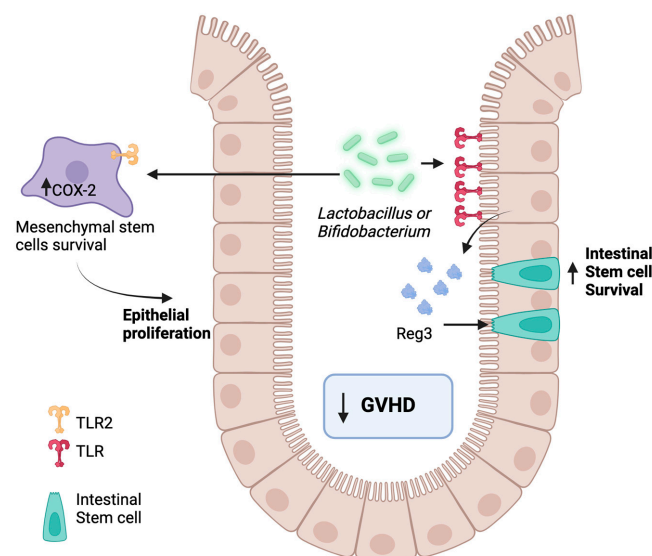


Figure 2. Activation of Toll-like receptors (TLRs) by the microbiome reduces GVHD by preserving the integrity of the intestinal epithelium. Conditioning treatments induce changes in TLR expression profiles. The activation of the TLR by bacteria like *Lactobacillus* or *Bifidobacterium* triggers survival and proliferative signals via the Cox-2 or Reg3 pathways into mesenchymal and intestinal stem cells, respectively, thereby enhancing and maintaining tissue integrity and mitigating GVHD. 5-FU, 5-fluorouracil; MTX, methotrexate; DAMPs, danger-associated molecular patterns; Reg3, regenerating family member 3; COX-2, cyclooxygenase-2. Created in [BioRender.com](https://www.biorender.com).

6.3. Inflammasome and Microbiome During GVHD

Proinflammatory cytokines such as tumor necrosis factor- α (TNF- α) and interleukine-1 (IL-1), released by host tissues during conditioning treatment, in combination with interferon-gamma (IFN- γ) released by donor T cells, play a major role in enhancing GVHD [108]. However, the regulation of these cytokines by the microbiome in GVHD is controversial, especially the balance between proinflammatory IL-1 β /IL-18.

The pathogen-associated molecular patterns (PAMPs) and the antimicrobial peptides (AMPs) activate the inflammasome superfamily of leucine-rich repeats of the nucleotide-binding domain (NLR). The lipopolysaccharide (LPS) usually activates the sensor proteins NLRP3 in the canonical pathway during GVHD [109]. This triggers the sensor protein oligomerization driving ASC recruitment and polymerization into the ASC speck structure. The CARD domain of the ASC recruits caspase-1 monomers via the CARD–CARD interaction, unleashing the protease function. The dimerization of caspase-1 and autocleavage allows the conversion of pro-IL-1 β to the mature form of IL-1 β and pro-IL-18, IL-18 [110]. The proinflammatory IL-1 β is associated with worse GVHD outcomes [111]; however, its blockage does not mitigate GVHD, indicating that IL-1 β production is not a determining factor per se in the development of GVHD [112].

On the other hand, the administration of IL-18 in GVHD mouse models decreases histopathology and results in significantly improved survival through the enhanced Fas-mediated apoptosis of donor T cells [113]. Additionally, the role of IL-18 depends on the GVHD context; in CD4+-dependent and MHC class II-disparate models, IL-18 regulates the GVHD. However, it aggravates GVHD in CD8+-dependent and MHC class I-disparate models [114]. In humans, the administration of IL-18 is associated with Th2 polarization instead of the characteristic Th1 profile in GVHD [115]. This suggests that the activation of the inflammasome by certain microorganisms, and the subsequent production of IL-18, can alter the T cell response during transplantation, shifting the balance toward a Th2 response rather than the typical Th1 response associated with GVHD [82].

SCFAs produced by the gut microbiota (GM) modulate leukocyte activity and suppress the release of inflammatory cytokines [116]. In the context of the microbiome and GVHD, it has been reported that microbial metabolites such SCFAs, butyrate, and propionate mitigate GVHD by G-protein-coupled receptor 43 (GPR43)—NLRP3 inflammasome pathway increasing IL-18 production [117]. Nevertheless, the NLRP3 pathway is not the only one that can contribute to IL-18 microbiome-mediated production.

NLRP6, another IL-18 inducer predominantly expressed in the intestine, is postulated to play an important role in maintaining microbiome homeostasis [118]. Evidence shows that *Asc*^{-/-}, *Caspase-1*^{-/-}, *Nlrp6*^{-/-}, and *Il18*^{-/-} mice, but not *Il1b*^{-/-} or *Il1R*^{-/-} mice, develop severe colitis after dextran sodium sulfate (DSS) treatment. Additionally, wild-type mice co-hosted with these modified mice were also more susceptible to DSS treatment, indicating that the microbiota shared between the wild type and the *Asc*^{-/-}, *Caspase-1*^{-/-}, *Nlrp6*^{-/-}, and *Il18*^{-/-} mice directly impact DSS-induced colitis [119].

In addition to the previously mentioned evidence, it has been shown that single nucleotide polymorphisms (SNPs) in receptors such as NOD2/CARD15 can lead to a reduction in nuclear factor kappa B (NF- κ B) pathway activation and the subsequent release of proinflammatory cytokines like IL-18. This, in turn, increases mortality and the incidence of graft-versus-host disease (GVHD) following hematopoietic stem cell transplantation (HSCT) [120]. Moreover, the risk associated with this mutation can be reduced depending on the type of antibiotic used in different protocols, suggesting that the persistent microbiota may influence the risk factors associated with these SNPs, showing that inflammasome activation is needed, to some degree, to improve the GVHD [121].

The evidence previously mentioned indicates that the NLRP pathway can be influenced by the microbiota and has an impact on GVHD outcomes through the IL-18 signal, potentially mitigating the disease (Figure 3). Additionally, the induction of IL-18 does not appear to impact the graft-versus-leukemia (GvT) effect [115]. However, the optimal balance of bacteria that can stimulate the IL-18 response needs to be further explored to enhance the management of dysbiosis during hematopoietic stem cell transplantation.

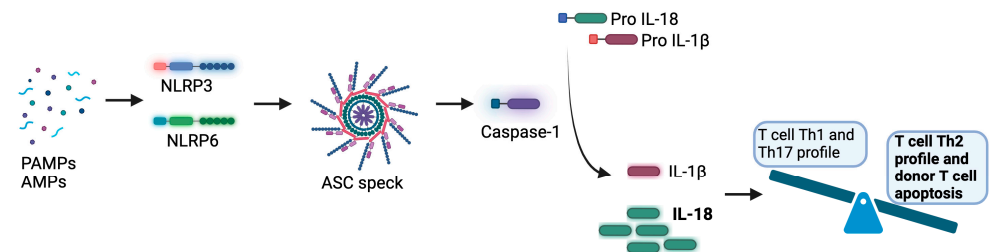


Figure 3. Inflammasome pathway activation by PAMPs and AMPs in a GVHD context. The activation of NLRP3 and NLRP6 by PAMPs and AMPs from the microbiota leads to an increase in IL-18 that changes the T cell profile from Th1/Th17 to Th2 T, with an increase in donor T cell apoptosis responsible for the development of GVHD. Created in [BioRender.com](#).

7. Prevention and Treatment

Due to the findings from animal models and observational studies, the microbiota and the intestinal inflammatory environment are now considered potential therapeutic targets in the field of hematopoietic transplantation (Figure 4).

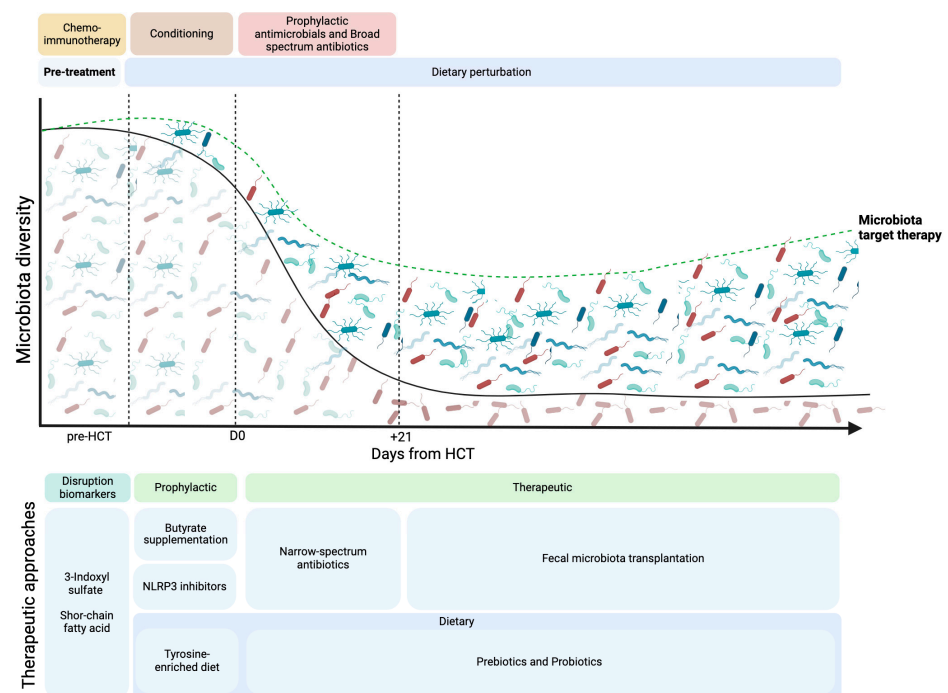


Figure 4. Dynamic changes in the microbiota profile occur during hematopoietic transplantation, with further alterations induced by treatments targeting dysbiosis. During the transplantation process, a reduction in microbiota diversity is commonly observed, particularly during the broad-spectrum antibiotic use phase to prevent infections during the aplasia period. This reduction in diversity can lead to the proliferation of bacteria associated with worsened GVHD outcomes. Treatment strategies aimed at restoring microbiota heterogeneity could increase the abundance of “beneficial” bacteria, which may positively impact GVHD severity in patients undergoing hematopoietic stem cell transplantation. Created in [BioRender.com](#).

Efforts have been made to identify the biomarkers of microbiota disruption and their impact on GVHD. Among the most studied are 3-indoxyl sulfate (3-IS) in urine and fecal SCFAs. 3-IS is a byproduct of indole metabolism produced by commensal gut bacteria and excreted in the urine. Low urinary levels of 3-IS have been described in transplant patients undergoing antibiotic treatment and in cohorts with gut GVHD, suggesting it as a biomarker for microbiota diversity. Additionally, 3-IS has been shown to modulate intestinal barrier function and the expression of proinflammatory genes in intestinal epithelial cells [122]. The indole derivatives activate the aryl hydrocarbon receptor (AhR), inducing IL-22 production [123]. This cytokine can activate the Reg3 γ pathway mentioned earlier, serving as a survival signal for intestinal stem cells and Paneth cells [105,106]. Additionally, the indole derivatives can mitigate GVHD by protecting the gut from radiation-induced tissue damage during myeloablative conditioning. This occurs through the induction of the type I interferon response, which increases epithelia repair and barrier integrity, limiting bacterial translocation [124].

Another relevant group of microbial metabolites are SCFAs (butyrate, propionate, and acetate). SCFAs act locally in the gastrointestinal tract by promoting mucus secretion, strengthening epithelial integrity, inducing the release of anti-inflammatory cytokines such as IL-10 and IL-18, promoting Treg differentiation by blocking histone deacetylases, and facilitating the acetylation of histone H3 in the Foxp3 locus of Tregs [80,81]. A decrease in fecal SCFAs has been observed in patients with aGVHD, and low blood levels of butyrate and propionate have been seen in patients with cGVHD [125]. In murine models, butyrate supplementation reduced the severity of GVHD [126].

Based on the absence of GVHD in decontaminated murine models [34], various decontamination approaches have been previously attempted in humans with controversial results [28,127]. Currently, strategies have been proposed to modify antibiotic use by selecting more narrow-spectrum antimicrobials (e.g., cefepime or rifaximin), which have less impact on beneficial anaerobic bacteria (such as Chloridoids), thus reducing antibiotic-associated dysbiosis [128]. There is also a focus on reducing the duration of broad-spectrum antibiotics and exploring new beta-lactams that could protect the gut microbiome [129].

Diet is one of the most influential factors shaping the composition of the microbiota, yet it remains one of the least explored areas in transplantation. Murine studies have shown that a tyrosine-enriched diet administered prior to HSCT promotes positive changes in the microbiome and reduces aGVHD [130], while diets rich in choline have the opposite effect, increasing GVHD rates [131]. In humans, the benefits of a specific diet aimed at modifying the microbiome are not yet clear; however, various studies have demonstrated the advantages of enteral nutrition over parenteral nutrition in terms of maintaining intestinal microbiota diversity [132].

Among the most promising and continuously evolving strategies are those involving prebiotics and probiotics. Prebiotics are substances that act as nutrients for the microbiota, typically oligosaccharides that commensal bacteria metabolize to produce SCFAs and other metabolites with beneficial immunomodulatory effects. Studies on the supplementation with resistant starch and GFO (glutamine, fiber, and oligosaccharide) have shown a reduction in the incidence and severity of aGVHD while preserving butyrate-producing bacteria [133]. However, the benefits of prebiotics appear to be transient, and their effectiveness may be limited [134].

Probiotics are microorganisms linked to multiple beneficial effects for humans, generally present in yogurt, some cheeses, and fermented food, and their commercially available formulation mainly contains *Lactobacillus* and *Bifidobacterium* species. This therapeutic approach has been difficult to evaluate in immunocompromised patients, as several cases of systemic infections associated with over-the-counter probiotics have been

reported [135,136]. There have been some retrospective studies on *Lactobacillus* species consumption before day +100 post-HSCT and the presentation of any adverse events [137,138]. Recently, an interventional study used a symbiotic mixture containing high levels of seven safe bacterial strains plus fructo-oligosaccharides in 21 allo-HSCT patients during 21 days prior transplantation, and it found the absence of any aGVHD signs (0% vs. 25% $p = 0.047$), higher Treg population on day 28 (2.54% vs. 1.73% $p = 0.01$), and higher OS (90% vs. 75% $p = 0.234$) in symbiotic therapy vs. control samples [139].

Fecal microbiota transplantation (FMT) is based on restoring microbiota diversity by directly reintroducing microorganisms to restore microbiome homeostasis. Although the most significant application of FMT has been in treating recurrent *Clostridium difficile* infection, the role of the microbiota in various systemic diseases (such as inflammatory bowel disease, irritable bowel syndrome, metabolic syndrome, eradication of multidrug-resistant organisms, and some autoimmune diseases) has broadened its applicability [140–142].

In the area of HSCT, some clinical case series and small prospective studies have shown complete remissions of GVHD, even in patients with steroid-refractory or steroid-dependent disease [143,144]. Taur and colleagues conducted a randomized controlled trial using autologous FMT derived from a stool bank of 14 HSCT patients, and compared to controls, the autoFMT group achieved a restoration of their intestinal microbiota composition to the pre-antibiotic and pre-transplantation baseline [145]. In the FMT2017002 trial (NCT03148743), 23 patients with grade IV steroid-refractory gastrointestinal GVHD received FMT via a nasojunal or gastric tube. On day 14 after FMT administration, 12 patients in the treatment group (52%) achieved permanent remission, compared to none in the control group. Additionally, the treatment group showed significantly better overall survival (OS) rates (>539 days vs. 107 days, HR 4.4, 95% CI, 1.5–13.04; $p = 0.008$), with no major adverse effects [146].

In a randomized double-blind phase II trial involving patients with acute myeloid leukemia and recipients of allogeneic hematopoietic stem cell transplantation (FMT group $n = 49$, placebo group $n = 25$), participants were treated with standardized oral encapsulated FMT versus placebo after neutrophil recovery. FMT was administered following each course of antibacterial antibiotics. In the allo-HSCT cohort, the rate of infections was lower in the FMT group, although this was not statistically significant. The FMT group also showed a restoration of microbiota diversity post-antibiotics and a reduction in the dominance of certain microorganisms, such as *Enterococcus*, *Streptococcus*, *Veillonella*, and *Dialister*. However, the cumulative incidence of GVHD (grade II–IV) was higher in the treatment group (29.8% vs. 8.3%, $p = 0.05$) [147].

Despite the promising potential for FMT, one of the biggest concerns is the safety of this therapy in immunocompromised patients, who also have altered intestinal barriers and a high risk of bacterial translocation. Many studies have emphasized the safety of FMT when performed after neutrophil engraftment, and in many cases, FMT was administered closer to day 100 post-transplantation, showing benefits in cGVHD or late-onset aGVHD. Other areas under investigation include donor screening protocols, selection criteria, optimizing the preservation of obligatory anaerobic organisms, identifying the best administration routes (capsules, nasoduodenal infusion, and enema), and determining the appropriate doses needed to achieve clear, lasting benefits in HSCT patients.

Although a healthy microbiota and the induction of IL-18 play a pivotal role in mitigating GVHD [118,119], both the microbiota and DAMPs during the myeloablative conditioning phase may have an inverse effect in amplifying the initial stage of GVHD [148–150]. A phase 2 trial using CD24Fc to reduce inflammation from DAMPs during the conditioning phase of bone marrow transplantation found increased GVHD-free survival compared to controls [151]. This result suggests the relevance of targeting inflammation during this

specific window of HSCT for future therapies. Although no NLRP3 inhibitors have been approved for use in humans, several molecules—including DFV890, ZYIL1, dapasuntrile, and GDC-2394—are currently being tested in early clinical trials [152–156]. NLRP3 inhibitors have the potential to be used as prophylaxis prior to conditioning treatment, in combination with other therapies aimed at improving the microbiota. However, the prolonged inhibition of NLRP3, in the context of GVHD, should be approached with caution, as some studies suggest that NLRP3 is not only essential for the release of IL-18 but also plays a critical role in the generation and function of the immune regulatory myeloid-derived suppressor cells (MDSCs), which contribute to controlling inflammation in this context [157,158].

Currently, there is no targeted therapy for the microbiota in transplant patients or those with GVHD that have completed phase III trials or have been approved by regulatory agencies. However, multiple clinical trials are underway, aimed at modifying nutritional support (gluten-free diet NCT03102060, intelligent nutritional support NCT05590091, high dose vitamin A NCT03719092), probiotic administration (*Clostridium butyricum* MIYAIRI 588 NCT01010867), prebiotics (resistant starch NCT05135351, inulin NCT04111471, prebiotic-rich food NCT04629430), the optimization of antibiotics (Rifaximin prophylaxis NCT03529825, antibiotic optimization NCT03727113), and fecal microbiota transplantation (prophylaxis of aGVHD NCT06026371; steroid-resistant/dependent aGVHD treatment NCT03214289, NCT03812705, NCT03359980, NCT03819803, NCT04139577; prophylaxis of *C. difficile* NCT02269150). Recently, pooled allogenic fecal matter from screened donors (MaaT013) has been developed for use in patients with GI-aGVHD, and the early results from the ARES trial (NCT04769895) appear promising.

8. Conclusions

The outcome of allo-HSCT is likely influenced by the complex interplay between various bacterial species, highlighting the critical role of microbiota ecology. The disruption of this balance during hematopoietic transplantation can impair colonization resistance, promoting the overgrowth of pathogenic microbes and complicating patient outcomes. While an increase in microbial diversity has been consistently linked to mitigating GVHD and mortality, the specific contributions of bacterial groups such as Enterococci and Lactobacilli remain under investigation. Although current data are limited, emerging evidence suggests that the microbiota may function as both a marker and a modulator of GVHD risk by maintaining gastrointestinal tissue integrity and modulating the donor T cell response. However, given the complexity of these interactions, further studies are essential to define the composition of beneficial microbiota and their mechanisms of action in allo-HSCT recipients.

With the rapid advancement of molecular techniques, the integration of multi-omics technologies, and more sophisticated bioinformatic platforms, microbiome analysis could soon be incorporated into the routine assessments for HSCT recipients. This would allow for the assessment of gut ecosystem function and the detection of potential biomarkers to prevent GVHD, as well as the identification of bacterial populations associated with adverse outcomes. Such insights may enable microbiota replenishment through interventions like fecal microbiota transplants or the administration of off-the-shelf microbial communities to restore a healthy microbiome.

Author Contributions: Conceptualization, P.P.-L. and H.G.-B.; writing—original draft preparation, P.P.-L., H.G.-B., S.M.-M. and J.C.Z.; writing—review and editing, P.P.-L., H.G.-B., S.M.-M. and J.C.Z.; visualization, P.P.-L. and H.G.-B. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created.

Conflicts of Interest: Authors Hernando Gutierrez-Barbosa, Sandra Medina-Moreno and Juan C Zapata were employed by the company Viriom Inc. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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